

# **SOCIOL 723: Social Statistics II**

## Week 4, Maximum Likelihood: Applications

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# Today

Putting  $X$  Inside  $\theta$

Binary Outcomes

Outcomes With Several Categories

Count Outcomes

Multilevel Models

Stars: ★ examinable; ★ never examined.

## X inside theta

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## Where Week 3 Left Us ★

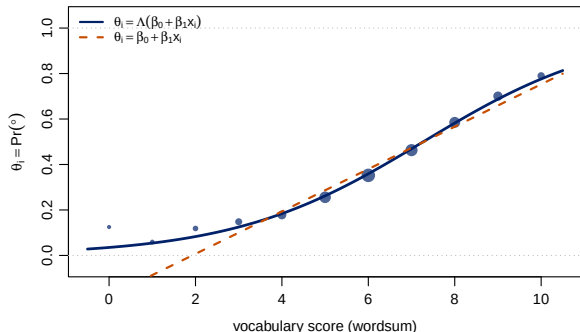
One parameter  $\theta$ , one curve  $\ell(\theta)$ , and everything read off that curve:

- the **peak** is the estimate  $\hat{\theta}$ ;
- the **curvature** at the peak, i.e., the second-order derivative, is the information; one over its square root is the standard error;
- a null  $\theta_0$  is tested by how far it is from the peak: the **gap** in SE units (Wald,  $z^2$ ) or the **drop** in  $\ell$ , doubled (LR). On a parabola they agree; when they disagree, believe LR;
- models are compared by LR when nested, by BIC when not.

**Today.**  $\theta$  was the same for everyone. Regression gives each person their own  $\theta_i$ , built from  $X_i$ : keep Week 3's likelihood, substitute  $\theta_i = h(X_i'\beta)$ , and estimate, standard errors, and tests carry over. The new work: choosing  $h$  and reading the result.

# One $\theta$ for Everyone Is Not a Regression ★

Week 3's Bernoulli gives every respondent the same  $\theta$ , the share with a degree. A regression lets  $\theta_i$  depend on  $X_i$ . First try: Week 1's line,  $\theta_i = \beta_0 + \beta_1 X_i$ .



GSS: share holding a bachelor's degree by vocabulary score (dot size = number of respondents).

- The line reaches  $\theta_i < 0$  below a score of 2. With four covariates, 140 of 3,509 fitted “probabilities” fall outside  $[0, 1]$ .
- The dots bend. The line cannot.

For a rate the same problem:  $\theta_i = X_i' \beta$  goes negative, and a Poisson with a negative rate is not a distribution.

## The Fix: a Function With the Right Range ★

Keep the line  $X_i'\beta$  and pass it through a function whose *output* lives where  $\theta$  must:

| Week 3 model           | its $\theta$ lives in | $\theta_i = h(X_i'\beta)$                                        | the regression     |
|------------------------|-----------------------|------------------------------------------------------------------|--------------------|
| Normal, mean $\mu$     | $(-\infty, \infty)$   | $\mu_i = X_i'\beta$                                              | OLS (the hook)     |
| Bernoulli, prob. $p$   | $(0, 1)$              | $p_i = \Lambda(X_i'\beta) = e^{X_i'\beta} / (1 + e^{X_i'\beta})$ | logit              |
| Bernoulli, prob. $p$   | $(0, 1)$              | $p_i = \Phi(X_i'\beta)$                                          | probit             |
| Poisson, rate $\theta$ | $(0, \infty)$         | $\theta_i = e^{X_i'\beta}$                                       | Poisson regression |

- $\Lambda$  and  $\Phi$  are S-curves from 0 to 1 (the logistic and normal CDFs);  $e^{(\cdot)}$  is positive.
- The **link function** is the inverse,  $g = h^{-1}$ , mapping  $\theta$  back to the line:  
 $g(p) = \log \frac{p}{1-p}$ ,  $g(\theta) = \log \theta$ . Textbooks start from  $g$ ; think in  $h$ .
- Every row is a *generalized linear model*: a Week 3 distribution, a line, and an  $h$  between them.

## What Each Regression Is Doing, in Week 3's Terms ★

Substitute  $\theta_i = h(X_i'\beta)$  into Week 3's log-likelihood:

$$\ell(\beta) = \sum_{i=1}^n \log f(y_i \mid \theta_i), \quad \theta_i = h(X_i'\beta).$$

- **Estimate:** the peak, in  $k$  dimensions. No closed form except OLS; Newton–Raphson, which `glm()` runs in a few steps.
- **Standard errors:** the curvature at the peak, inverted. `summary()` prints the square roots.
- **Tests:** `summary()`'s  $z$  is Wald; `anova()` is LR.
- **Comparison:** AIC and BIC from  $\ell(\hat{\beta})$ .

The only new work is interpretation.  $\beta_j$  moves the line; how much it moves  $\theta_i$  depends on  $h$  and on where person  $i$  sits.

# Binary

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# The Logit Model, Written Two Ways ★

Two ways to write the same model, the textbook's and the last section's:

$$\log \frac{p_i}{1-p_i} = X_i' \beta$$

textbook form: log-odds linear in  $X$

$$\frac{p_i}{1-p_i} = e^{X_i' \beta}$$

exponentiate both sides

$$p_i = e^{X_i' \beta} - e^{X_i' \beta} p_i$$

multiply by  $1 - p_i$

$$p_i (1 + e^{X_i' \beta}) = e^{X_i' \beta}$$

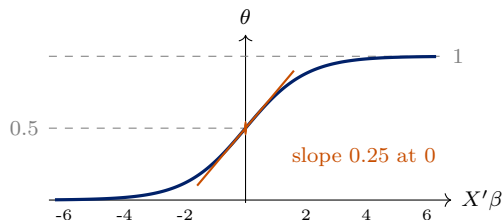
collect the  $p_i$  terms

$$p_i = \frac{e^{X_i' \beta}}{1 + e^{X_i' \beta}} = \Lambda(X_i' \beta)$$

divide: probability form

- Every step reverses: the forms are equivalent, and  $\Lambda$  is the inverse of the log-odds. Read coefficients on the first line ( $\beta_j$  adds to the log-odds,  $e^{\beta_j}$  multiplies the odds); compute on the last.
- Also from line 2:  $1 - p_i = 1/(1 + e^{X_i' \beta})$ , used next.

# The S-Curve ★



- $\Lambda$  is an S-curve: flat near 0 and 1, steepest at  $X'\beta = 0$  where  $\theta = 0.5$  (slope 0.25). Any real  $X'\beta$  lands inside  $(0, 1)$ .
- The slope at a person's  $X'\beta$  is  $\theta(1 - \theta)$ : one unit of  $X'\beta$  moves the probability by up to 0.25 in the middle and almost nothing in the tails.
- Only 0s and 1s are observed on the vertical axis; the likelihood estimates the curve from them.

## The Logit Model, and One Fitted to the GSS ★

Week 3's Bernoulli log-likelihood with  $p_i = \Lambda(X_i'\beta)$  substituted:

$$\ell(\beta) = \sum_{i=1}^n \left[ y_i \log p_i + (1 - y_i) \log(1 - p_i) \right] = \sum_{i=1}^n \left[ y_i X_i' \beta - \log(1 + e^{X_i' \beta}) \right]$$

(second form:  $\log \frac{p_i}{1-p_i} = X_i' \beta$ ,  $\log(1 - p_i) = -\log(1 + e^{X_i' \beta})$ ; concave, so the peak is unique when it exists.) Holding a bachelor's degree,  $n = 3,509$ :

|                            | $\hat{\beta}$ | SE    | $z$   | $e^{\hat{\beta}}$ |
|----------------------------|---------------|-------|-------|-------------------|
| intercept                  | -5.81         | 0.25  | -23.7 |                   |
| parents' education (years) | 0.207         | 0.013 | 15.4  | 1.23              |
| vocabulary score (0-10)    | 0.391         | 0.024 | 16.3  | 1.48              |
| female                     | 0.279         | 0.078 | 3.6   | 1.32              |
| black                      | -0.114        | 0.110 | -1.0  | 0.89              |

Peak, curvature, Wald ruler: all Week 3. The last column is the interpretation problem.

## Newton by Hand on the GSS: Degree on Vocabulary

Week 3's Newton step, with a gradient and a curvature matrix:

$$\beta^{[t+1]} = \beta^{[t]} + \mathcal{I}(\beta^{[t]})^{-1} S(\beta^{[t]}), \quad S(\beta) = \sum_i (y_i - p_i) X_i, \quad \mathcal{I}(\beta) = \sum_i p_i(1 - p_i) X_i X_i'.$$

Start at  $\beta = (0, 0)$ , so every  $p_i = 0.5$ :

| step $t$ | $\beta_0$ | $\beta_1$ | $\ell$   | score $S = (S_0, S_1)$ |
|----------|-----------|-----------|----------|------------------------|
| 0        | 0         | 0         | -2432.3  | (-313.5, -792.5)       |
| 1        | -2.718    | 0.373     | -2133.4  | (-38.8, -82.4)         |
| 2        | -3.272    | 0.450     | -2125.6  | (-2.4, -5.1)           |
| 3        | -3.313    | 0.456     | -2125.53 | (0.0, 0.0)             |

- Three steps and the score is zero: the peak. `glm(ba ~ wordsum)` reports  $(-3.313, 0.456)$ .
- The score says which way is uphill, the curvature matrix how far. Inverted at the peak, it is the variance matrix.

# The Logit Score Is the OLS Normal Equation

Differentiate, using  $\partial\Lambda(z)/\partial z = \Lambda(z)(1 - \Lambda(z))$ :

$$\frac{\partial \ell}{\partial \beta} = \sum_{i=1}^n \left[ y_i X_i - \frac{e^{X_i' \beta}}{1 + e^{X_i' \beta}} X_i \right] = \sum_{i=1}^n (y_i - p_i) X_i \stackrel{!}{=} \mathbf{0}.$$

## Read it

$\sum_i (y_i - \hat{p}_i) X_i = \mathbf{0}$ : the residuals  $y_i - \hat{p}_i$  are orthogonal to the regressors, **exactly the OLS normal equations** with the fitted probability in place of the fitted value. The fitted probabilities average to the sample share, 0.41. (OLS with an intercept has the same property:  $\bar{\hat{y}} = \bar{y}$ .)

No closed form: Newton, as on the last frame. The curvature

$\mathcal{I}(\beta) = \sum_i p_i(1 - p_i) X_i X_i'$ , inverted at  $\hat{\beta}$ , is the variance matrix: the SE column.

## Three Scales, One Coefficient ★

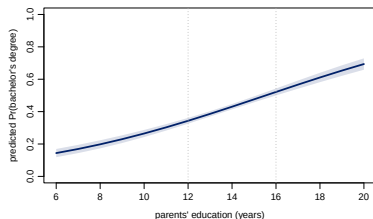
Take vocabulary:  $\hat{\beta} = 0.391$ .

| Scale       | Quantity                       | $\hat{\beta} = 0.391$ says          |                       |
|-------------|--------------------------------|-------------------------------------|-----------------------|
| Log-odds    | $\log \frac{p}{1-p} = X'\beta$ | one more word adds 0.391            | linear, not intuitive |
| Odds        | $\frac{p}{1-p} = e^{X'\beta}$  | odds multiply by $e^{0.391} = 1.48$ | the <i>odds ratio</i> |
| Probability | $p = \Lambda(X'\beta)$         | depends on where you start          | what we care about    |

- An odds ratio of 1.48 fits  $0.05 \rightarrow 0.07$  and  $0.40 \rightarrow 0.50$  alike. It is not a risk ratio and not a difference in probabilities; do not read “48% more likely.”
- On the probability scale one more word adds  $\Lambda(\eta + 0.391) - \Lambda(\eta)$ : large near  $p = 0.5$ , tiny near 0 or 1.
- Report on the probability scale, and say *for whom*.

## Start With Predicted Probabilities ★

How likely is a degree if your parents finished high school (12 years) rather than college (16)? For people like those in the data:



| parents' educ | Pr(degree) | 95% CI       |
|---------------|------------|--------------|
| 8             | 0.20       | [0.17, 0.22] |
| 12            | 0.34       | [0.33, 0.36] |
| 16            | 0.52       | [0.50, 0.54] |
| 20            | 0.69       | [0.66, 0.73] |

- **Observed-value approach** (Hanmer and Kalkan 2013): set parents' education to 12 for *everyone*, keep every other covariate at its actual value, compute  $\hat{p}_i$ , average; repeat at 16. High school to college: 18 points.
- The standard presentation in sociology journals: a curve with a band across the key variable, everything else at observed values; coefficients in the appendix.  
`avg_predictions()`.

## Reading a Logit Slowly: Pager (2003, *AJS*) ★

“The Mark of a Criminal Record.” Matched testers applied for 350 jobs; the outcome is a callback (0/1); the regressors are *record*, *Black*, and their product. Appendix table B1:  $\hat{\beta}_{\text{record}} = -0.99$ ,  $\hat{\beta}_{\text{Black}} = -1.25$ , interaction  $-0.29$ . The base cell is white, no record, so  $\hat{\beta}_0 = -0.66$ : odds  $e^{-0.66} = 0.52$ , probability  $0.52/1.52 = 0.34$ , Pager’s 34%.

| cell             | log-odds $X'\hat{\beta}$             | odds $e^{X'\hat{\beta}}$ | $\hat{p} = \text{odds}/(1 + \text{odds})$ | Pager |
|------------------|--------------------------------------|--------------------------|-------------------------------------------|-------|
| white, no record | $-0.66$                              | 0.52                     | 0.34                                      | 34%   |
| white, record    | $-0.66 - 0.99 = -1.65$               | 0.19                     | 0.16                                      | 17%   |
| Black, no record | $-0.66 - 1.25 = -1.91$               | 0.15                     | 0.13                                      | 14%   |
| Black, record    | $-0.66 - 0.99 - 1.25 - 0.29 = -3.19$ | 0.04                     | 0.04                                      | 5%    |

Read one column at a time. A coefficient *adds* on the log-odds scale (column 2); exponentiating turns adding into multiplying (column 3); odds over one plus odds is a probability (column 4), which is what the audit counted (column 5; the small gaps are Pager’s two controls).

## The Average Marginal Effect, Person by Person ★

Person  $i$ 's slope in vocabulary is  $\hat{p}_i(1 - \hat{p}_i) \times 0.391$ . Four real respondents, then the average over all 3,509:

| respondent                        | $X'_i \hat{\beta}$ | $\hat{p}_i$ | $\hat{p}_i(1 - \hat{p}_i)$ | slope = $\hat{p}_i(1 - \hat{p}_i) \times 0.391$ |
|-----------------------------------|--------------------|-------------|----------------------------|-------------------------------------------------|
| parents 12, voc. 1, male, white   | -2.94              | 0.05        | 0.048                      | 0.019                                           |
| parents 12, voc. 6, male, Black   | -1.10              | 0.25        | 0.188                      | 0.073                                           |
| parents 13, voc. 8, male, white   | 0.01               | 0.50        | 0.250                      | 0.098                                           |
| parents 20, voc. 8, female, white | 1.73               | 0.85        | 0.128                      | 0.050                                           |
| average over all 3,509            |                    |             | 0.192                      | <b>0.075</b> = AME                              |
| one person at the sample means    | -0.49              | 0.38        | 0.236                      | 0.092 = MEM                                     |

- Same 0.391 in every row; what differs is where each person sits on the S-curve: the slope peaks at  $\hat{p} = 0.5$  and is a fifth of that at  $\hat{p} = 0.05$ .
- The AME is the average of the last column, equally  $0.192 \times 0.391$ . The MEM plugs in the means (13.2 years, 6.3 words, 0.48 female, 0.15 Black): a person who does not exist, near the steep part.

## Then Marginal Effects: One Number per Variable ★

Predicted probabilities take a figure per variable. A table wants one number per variable, comparable across models and papers: the **average slope** of that curve.

- For one person the slope of the probability in  $X_j$  is  $\partial p_i / \partial X_{ij} = p_i(1 - p_i) \beta_j$ :  $\beta_j$  scaled by where the person sits on the S-curve (0 to 0.25).
- **Average marginal effect (AME)**, 0.075: “averaged over the people in the sample, one more word raises the probability of a degree by 7.5 percentage points.” The last frame’s average.
- **Marginal effect at the mean (MEM)**, 0.092: “for one hypothetical person at the mean of every covariate, one more word adds 9.2 points.” Older papers report it; report the AME.

`avg_slopes(m)` computes all of them.

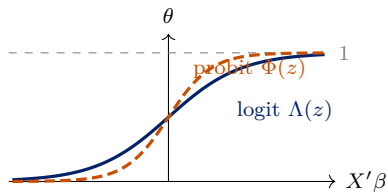
## Uncertainty for These Quantities

An AME or a predicted probability is a *nonlinear* function of  $\hat{\beta}$ ; its SE is not in the regression table. Three ways:

1. **Delta method:** linearize  $g(\hat{\beta})$  and use  $\nabla g' \hat{V}(\hat{\beta}) \nabla g$ . Analytic, fast, the default in `marginalEffects`. The SEs on the last two frames are these.
2. **Simulation** (King, Tomz, and Wittenberg 2000): draw  $\tilde{\beta}^{(s)} \sim \mathcal{N}(\hat{\beta}, \hat{V})$ , recompute the quantity at each draw, take percentiles. Any nonlinearity, no calculus.
3. **Bootstrap** (Week 2): resample rows, refit, recompute. Slowest, fewest assumptions.

**Do not** build an interval on the log-odds scale and transform its endpoints if what you want is an interval for a *difference* in probabilities.

## Probit: Another S-Curve ★



- Any function that rises from 0 to 1 can play  $h$ . The logistic  $\Lambda$  is one; the normal CDF  $\Phi$  is the other common choice:  
 $p_i = \Phi(X_i'\beta)$ , the **probit**.
- $\Phi$  is a slightly narrower S. Stretch its axis by 1.7 and it sits on  $\Lambda$  to within 0.01: same shape, different units.

|                      | parents' educ | vocabulary    | female        | black           |
|----------------------|---------------|---------------|---------------|-----------------|
| logit $\hat{\beta}$  | 0.207         | 0.391         | 0.279         | -0.114          |
| probit $\hat{\beta}$ | 0.117         | 0.233         | 0.159         | -0.070          |
| ratio                | 1.77          | 1.68          | 1.75          | 1.64            |
| AME, logit / probit  | 0.040 / 0.038 | 0.075 / 0.075 | 0.054 / 0.052 | -0.022 / -0.023 |

## Logit or Probit: a Recommendation ★

- **Default to logit.** It has the cleaner likelihood, odds ratios for the audiences that want them, and it is what every package and most readers expect.
- **Probit when the next step needs a normal:** selection models, bivariate probit, or a subfield convention. Probit comes from economics and psychometrics, where it arrived first.
- **Never compare raw coefficients across nested models or across groups,** in either model. Adding a covariate rescales all the coefficients, even ones whose effect is unchanged (Allison 1999; Mood 2010). Compare AMEs or predicted probabilities, which live on the probability scale.
- **Never agonize.** AMEs from the two agree to two decimals; the choice is a footnote.

## Interactions in Nonlinear Models ★

Two questions: should the product  $X_1X_2$  be in the model, and do the models with and without it agree on probabilities? What each can say about  $X_1$ 's effect across  $X_2$ :

|              | log-odds scale                                           | probability scale                                   |
|--------------|----------------------------------------------------------|-----------------------------------------------------|
| no product   | one slope, $\beta_1$ , forced equal across $X_2$         | differs across $X_2$ anyway, by $p_i(1 - p_i)$ only |
| with product | $\beta_1 + \beta_{12}X_2$ , the data choose $\beta_{12}$ | differs by $p_i(1 - p_i)$ and by $\beta_{12}$       |

- **Include the product** whenever the question is whether  $X_1$ 's effect differs by  $X_2$ . Without it the answer is fixed in advance: zero on the log-odds scale, whatever the S-curve gives on the probability scale (Ai and Norton 2003; Berry, DeMeritt, and Esarey 2010).
- **The two models agree** on probabilities only when  $\hat{\beta}_{12} \approx 0$ ; otherwise the no-product model has the wrong probabilities in the cells.
- **Report** the interaction as a difference in AMEs of  $X_1$  between the  $X_2$  groups, with an interval.  $\hat{\beta}_{12}$  is on the wrong scale for the question.

## Interactions: Real Output, Number by Number ★

Does parents' education matter more for women? Add the product: `ba ~ pareduc * female + wordsum + black`.

|                               | $\hat{\beta}$ | SE    | $z$   |
|-------------------------------|---------------|-------|-------|
| parents' educ                 | 0.197         | 0.018 | 10.9  |
| female                        | -0.012        | 0.377 | -0.03 |
| parents' educ $\times$ female | 0.021         | 0.027 | 0.79  |

| AME of parents' educ | no product | with product  |
|----------------------|------------|---------------|
| among men            | 0.039      | 0.037         |
| among women          | 0.041      | 0.043         |
| difference (SE)      | 0.002      | 0.006 (0.004) |

- **Log-odds scale:** a year of parents' education adds 0.197 for men and  $0.197 + 0.021$  for women. The product is not significant ( $z = 0.79$ ,  $LR = 0.6$ ).
- **No-product column:** women's AME is already larger, 0.041 against 0.039, with no product in the model. That gap is the S-curve alone: women sit where it is steeper.
- **With-product column:** the gap is 0.006 per year, SE 0.004. This answers the question; the answer is no. The two models nearly agree because  $\hat{\beta}_{12}$  is near zero. `avg_slopes(m, variables = "pareduc", by = "female")`.

## Several Categories

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## The Setup: One Outcome, Four Regions ★

**The outcome.** Where a respondent lives: Midwest, Northeast, South, West ( $J = 4$ ).  
**These categories have no order:** the West is not “more” than the South. That is what makes this the multinomial, not the ordered, case.

**Week 3's Bernoulli, generalized.**  $Y_i$  takes one of  $J$  values with probabilities  $p_{i1}, \dots, p_{iJ}$  adding to one:

$$\ell = \sum_{i=1}^n \sum_{j=1}^J \mathbf{1}[Y_i = j] \log p_{ij} \quad \text{one nonzero term per person; } J = 2 \text{ is Week 3's Bernoulli.}$$

**The regression.** Four probabilities per person from  $X_i$ . Pick a base, the Midwest, and write one logit for every other region *against the base*:

$$\log \frac{p_{ij}}{p_{i,\text{Midwest}}} = X_i' \beta_j, \quad j = \text{Northeast, South, West.}$$

Three lines, three coefficient vectors. The base gets none: its own line would read  $\log(p_{i,\text{base}}/p_{i,\text{base}}) = 0$ ; writing  $\beta_{\text{base}} = \mathbf{0}$  is a convention so one formula covers all four ( $e^0 = 1$ ).

## From $J - 1$ Log-Odds to $J$ Probabilities ★

The same algebra as the two-ways frame, with a sum in the denominator:

$$p_{ij} = p_{i,\text{base}} e^{X'_i \beta_j} \quad \text{exponentiate the } j\text{th line}$$

$$1 = \sum_j p_{ij} = p_{i,\text{base}} \left( 1 + \sum_{m \neq \text{base}} e^{X'_i \beta_m} \right) \quad \text{add up all } J; \text{ the base contributes } e^0 = 1$$

$$p_{i,\text{base}} = 1 / \left( 1 + \sum_m e^{X'_i \beta_m} \right) \quad \text{solve for the base}$$

$$p_{ij} = e^{X'_i \beta_j} / \left( 1 + \sum_m e^{X'_i \beta_m} \right) \quad \text{substitute back: the multinomial logit}$$

- Positive and summing to one by construction. With  $J = 2$  this is  $\Lambda(X'_i \beta)$ : the logit is the special case. Estimation is Week 3 again, Newton over  $(J - 1)k = 15$  coefficients; `nnet::multinom`.
- **IIA**: the odds of  $j$  versus  $m$  never involve a third category, so adding an option cannot change the ratio of two old ones.

## Multinomial Logit on the GSS: Where People Live ★

Base Midwest; each row is one line, shown as  $e^{\hat{\beta}}$  (odds multipliers),  $n = 3,509$ :

| versus Midwest | education (per year) | Black       | female | age (per year) |
|----------------|----------------------|-------------|--------|----------------|
| Northeast      | 1.08                 | 1.32        | 1.08   | 1.00           |
| South          | 0.98                 | <b>2.88</b> | 1.21   | 1.00           |
| West           | 1.01                 | 0.82        | 1.13   | 0.99           |

Reading one cell: a Black respondent's odds of living in the South *rather than the Midwest* are 2.88 times a white respondent's, other things equal. Change the base and every number changes, so report the four probabilities, observed-value style:

|           | Midwest | Northeast | South | West |
|-----------|---------|-----------|-------|------|
| not Black | 0.26    | 0.17      | 0.33  | 0.23 |
| Black     | 0.16    | 0.14      | 0.58  | 0.12 |

Education moves people toward the Northeast ( $0.14 \rightarrow 0.18$ , 12 to 16 years) and out of the South ( $0.40 \rightarrow 0.35$ ). All 12 slopes:  $LR = 171$ .

## How the Four Probabilities Are Computed ★

**Step 1: one person, one setting.** A respondent with 16 years of education, female, aged 40, Black *set* to 1. One linear predictor per non-base line, then the formula:

|                                                | Midwest (base) | Northeast | South | West   |
|------------------------------------------------|----------------|-----------|-------|--------|
| $X_i' \hat{\beta}_j$                           | 0              | 0.011     | 1.352 | -0.196 |
| $e^{X_i' \hat{\beta}_j}$                       | 1              | 1.011     | 3.867 | 0.822  |
| $\hat{p}_{ij} = e^{X_i' \hat{\beta}_j} / 6.70$ | 0.149          | 0.151     | 0.577 | 0.123  |

The denominator  $6.70 = 1 + 1.011 + 3.867 + 0.822$  is shared by the four cells, which is why they add to one. Set Black = 0 for the same person and the row becomes 0.243, 0.186, 0.327, 0.244.

**Step 2: everyone, then average.** Repeat step 1 for all 3,509 respondents with Black set to 1, each keeping their own education, gender, and age; average each column: 0.16, 0.14, 0.58, 0.12. Then with Black set to 0: 0.26, 0.17, 0.33, 0.23. Those are the last frame's two rows. `avg_predictions(m, variables = "black")`.

## Ordered Logit: Using the Order ★

Degree is ordered; a multinomial on it would spend 20 parameters ignoring that. Use the order: model the probability of being *at or below* each level,

$$\log \frac{\Pr(Y_i \leq j)}{\Pr(Y_i > j)} = \tau_j - X'_i \beta, \quad j = 1, \dots, J-1 \quad \implies \quad \Pr(Y_i \leq j) = \Lambda(\tau_j - X'_i \beta),$$

one  $\beta$  for every cut and  $J-1$  thresholds  $\tau_1 < \dots < \tau_{J-1}$ :  $4 + 4 = 8$  parameters.

Category probabilities are differences of neighbouring cumulative ones,

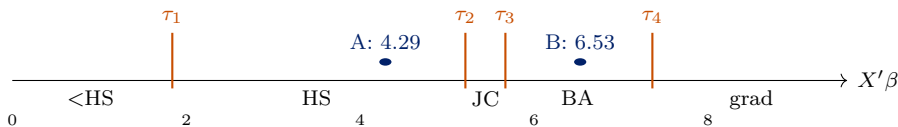
$p_{ij} = \Lambda(\tau_j - X'_i \beta) - \Lambda(\tau_{j-1} - X'_i \beta)$ . Larger  $X'_i \beta$  pushes probability up the ladder.

|                    | $\hat{\beta}$ | $z$  | cutpoint                    | $\hat{\tau}$ |
|--------------------|---------------|------|-----------------------------|--------------|
| parents' education | 0.208         | 19.9 | <HS   HS                    | 1.84         |
| vocabulary         | 0.358         | 18.6 | HS   junior college         | 5.21         |
| female             | 0.335         | 5.2  | junior college   bachelor's | 5.67         |
| black              | -0.024        | -0.3 | bachelor's   graduate       | 7.36         |

$\hat{\beta}$  reads like a logit coefficient; the cutpoints are the intercepts. `MASS::polr`.

## The Cutpoints Are Gates ★

Why  $J - 1$  extra parameters?  $\beta$  says how far a person sits along a line; the cutpoints say *where the category boundaries are* on that line. The slope alone cannot say how much of the line is “high school.”



$\Pr(Y_i \leq j) = \Lambda(\tau_j - X'_i\beta)$ : the chance of being at or below gate  $j$  is the S-curve at the *distance from the person to the gate*. Each category's probability is the mass between two gates; junior college is a narrow band, so it gets little mass for anyone.

|                                                | $X'\hat{\beta}$ | <HS  | HS   | JC   | BA   | grad |
|------------------------------------------------|-----------------|------|------|------|------|------|
| A: parents 12 yrs, vocabulary 5, male, white   | 4.29            | 0.08 | 0.64 | 0.08 | 0.16 | 0.04 |
| B: parents 16 yrs, vocabulary 8, female, white | 6.53            | 0.01 | 0.20 | 0.09 | 0.40 | 0.30 |

## Reading an Ordered Logit: Five Probabilities Move ★

A coefficient says the line moves; readers want the five probabilities. Observed-value predictions, vocabulary 4 versus 8:

|                | <HS   | HS    | junior college | bachelor's | graduate |
|----------------|-------|-------|----------------|------------|----------|
| vocabulary = 4 | 0.09  | 0.59  | 0.08           | 0.18       | 0.06     |
| vocabulary = 8 | 0.03  | 0.35  | 0.10           | 0.33       | 0.19     |
| difference     | -0.07 | -0.24 | +0.02          | +0.16      | +0.14    |

- Four more words move a quarter of the population out of “high school only,” mostly into bachelor’s and graduate degrees.
- The middle category barely moves: one  $\hat{\beta}$  pushes probability *through* a narrow band without piling it up. The sign of the effect on a middle category is not the sign of  $\hat{\beta}$ . (`avg_predictions()` again.)

## Proportional Odds, and Checking It ★

One  $\beta$  for every cutpoint is an assumption. Check it with four separate binary logits, “above cut  $j$ ” versus not:

| above ...          | <HS   | HS    | junior college | bachelor's |
|--------------------|-------|-------|----------------|------------|
| parents' education | 0.231 | 0.198 | 0.207          | 0.180      |
| vocabulary         | 0.407 | 0.348 | 0.391          | 0.331      |
| female             | 0.252 | 0.367 | 0.279          | 0.379      |

- The alternative: the same cumulative model with a separate  $\beta$  per cut, proportional odds nested by  $\beta_1 = \dots = \beta_4$  (not the multinomial, a different family).
- Brant test on the table: 32.5 on 12 df,  $p = 0.001$ . Vocabulary fails (12.3 on 3 df); the others pass. Its 0.36 averages 0.41 at the first cut and 0.28 at the rest.
- Remedy: free that slope only, partial proportional odds (`VGAM::vglm`, `ordinal::clm`).

## In the Literature: Multinomial Logit ★

**Kuo and Raley (2016, *Demography*)**. What happens to a first premarital cohabitation? Three outcomes in each month: it becomes a marriage, it dissolves, or it continues. A discrete-time multinomial logit on person-months from the National Survey of Family Growth, two cohorts.

- Within three years: 41% married and 35% separated in the 1990–95 cohort; 24% married and 41% separated in 2005–10.
- The retreat from marriage is educational: college-educated cohabitators' annual marriage probability held steady; for those without a degree it fell to less than half.
- Why not a logit on “married or not”: the alternative to marriage shifted from staying together to breaking up.

**Breen and Jonsson (2000, *ASR*)**. Swedish educational careers as multinomial transitions: “continue or leave” hides *which* branch a student continues into.

## In the Literature: Ordered Logit ★

**Zajacova, Huzurbazar, and Todd (2017, *Social Science & Medicine*).**

Self-rated health, five ordered categories from excellent to poor,  $N = 296,679$  adults in the National Health Interview Survey.

- Ordered logits by age group, with the female coefficient read as an odds ratio on the odds of *worse* health, the same at every cut.
- Ages 25–44: women's odds are 1.12 times men's with demographic controls only, and 1.00 once health conditions and socioeconomic status are added. Ages 65–84: 0.75, women healthier.
- The design is exactly this section's: one  $\beta$  per covariate, four cutpoints, and the same coefficient compared across models that add controls, which is where the proportional-odds check and the caution about comparing coefficients across models both matter.

# Counts

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## Poisson Regression: Week 3's Poisson With $\theta_i = e^{X_i'\beta}$ ★

Number of children,  $n = 3,509$ :  $Y_i \mid X_i \sim \text{Poisson}(\theta_i)$ ,  $\log \theta_i = X_i'\beta$ .

|                    | $\hat{\beta}$ | SE    | $e^{\hat{\beta}}$ | reading                      |
|--------------------|---------------|-------|-------------------|------------------------------|
| education (years)  | -0.045        | 0.005 | 0.956             | 4.4% fewer children per year |
| female             | 0.009         | 0.027 | 1.01              | no difference                |
| black              | 0.360         | 0.035 | 1.43              | 43% more                     |
| experience (years) | 0.012         | 0.001 | 1.01              | 1.2% more per year (age)     |
| ever married       | 0.947         | 0.043 | 2.58              | 2.6 times as many            |

- $e^{\hat{\beta}_j}$  multiplies the expected count, because  $h = \exp$  turns adding into multiplying; a small  $\hat{\beta}$  is about a percentage.
- **Exposure** ★: counts over different windows (years observed, population at risk) model the *rate*: `offset(log(exposure))`, a coefficient fixed at 1.
- Why not OLS on  $\log y$ :  $\log 0$  is undefined and 30% have no children. The likelihood uses the counts themselves. What can go wrong is the variance, next.

## Overdispersion: the Poisson's One Strong Claim ★

Week 3's Poisson has one parameter, so its variance is not free:

$\text{Var}(Y_i | X_i) = E[Y_i | X_i] = \theta_i$ . Real counts almost always vary more, because people differ in ways  $X$  does not capture: some are systematically more event-prone.

- Raw: mean 1.60, variance 2.11. Conditional on  $X$  the excess is smaller but there: `AER::dispersiontest` estimates  $\text{Var} = \theta_i(1 + 0.17)$ ,  $z = 3.96$ ,  $p < 0.0001$ .
- That test uses only the Poisson fit: it asks whether the residual variance grows with the mean faster than the Poisson allows, without fitting any alternative.
- **Consequence:** coefficients stay about right (the mean is still right), but **standard errors are too small**, here by about 7%, in other data by a factor of two or three.

Three repairs, next frame; all three leave  $\hat{\beta}$  nearly alone and fix the SEs.

## Three Repairs, Three Beliefs About the Variance

The Poisson standard error comes from the Poisson's variance claim,  $\text{Var}(Y_i | X_i) = \theta_i$ . Once that claim is false, the SE is wrong, and each repair replaces the claim with a different one:

| Repair            | Belief about $\text{Var}(Y_i   X_i)$                        | What changes                                |
|-------------------|-------------------------------------------------------------|---------------------------------------------|
| Quasi-Poisson     | $\phi \theta_i$ , one multiple everywhere                   | every SE, by one factor $\sqrt{\hat{\phi}}$ |
| Negative binomial | $\theta_i + \theta_i^2/\alpha$ , grows faster than the mean | the likelihood: $\hat{\beta}$ and SEs       |
| Robust (sandwich) | none; use the squared residuals as they are                 | SEs only, each on its own                   |

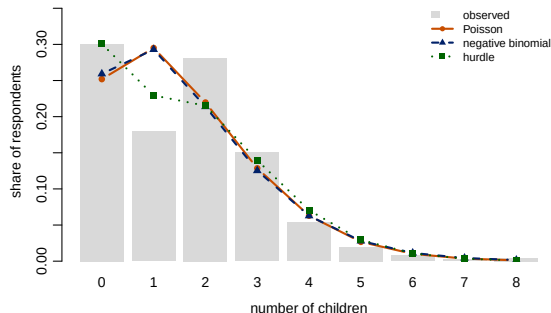
- All three leave  $\hat{\beta}$  essentially where the Poisson put it: the mean is not in question, only the scatter around it.
- They differ in how much they assume: one number, a particular shape, or nothing (paid for with noisier SEs).

Next: the arithmetic behind each number for one coefficient, *ever married*, whose Poisson SE is 0.0430.

## Where the Three Numbers Come From

- **Quasi-Poisson**, 0.0461.  $\hat{\phi} = \frac{1}{n-k} \sum_i (y_i - \hat{\theta}_i)^2 / \hat{\theta}_i = 1.149$ : under the Poisson each term averages 1; the counts scatter 15% more. Multiply every Poisson SE by  $\sqrt{1.149} = 1.072$ :  $0.0430 \times 1.072 = 0.0461$ . Nothing else moves.
- **Negative binomial**, 0.0437. A new likelihood with variance  $\theta_i + \theta_i^2/\alpha$ ; the data choose  $\hat{\alpha} = 31$ , so at the mean count 1.6 the variance is 1.68, not 1.60. Each observation's weight in the curvature falls by the factor  $1/(1 + \theta_i/\hat{\alpha}) \approx 0.95$ , and the SE grows by about 2%. Mild, because  $\hat{\alpha}$  is large.
- **Robust**, 0.0551. Week 2's sandwich,  $(X'WX)^{-1} [\sum_i X_i X_i' (y_i - \hat{\theta}_i)^2] (X'WX)^{-1}$ , uses each person's own squared residual. Why so much larger here: the excess scatter is not uniform. Among the never married the Pearson ratio is 1.66, among the ever married 0.96. One  $\hat{\phi} = 1.15$  averages them; the sandwich sees that this dummy compares exactly the two groups whose scatter differs.

## Excess Zeros ★



- Observed: 30% childless. Poisson predicts 25%, the negative binomial 26%. Neither has a way to make zeros special.
  - Sociologically zero *is* special: whether to have children and how many are different decisions. Two models that say so:
- **Hurdle:** a logit for zero versus positive, then a *truncated* count model for the positives. `pscl::hurdle`.
  - **Zero-inflated:** a mixture; a share of “structural zeros” who could never have a positive count, the rest a count model that can also produce zeros. `pscl::zeroinfl`.

## The Hurdle Model on the GSS ★

Same five covariates in both parts; count part Poisson.

|              | any children? (logit) |                   | how many, given some (Poisson) |                   |
|--------------|-----------------------|-------------------|--------------------------------|-------------------|
|              | $\hat{\beta}$         | $e^{\hat{\beta}}$ | $\hat{\beta}$                  | $e^{\hat{\beta}}$ |
| education    | -0.116                | 0.89              | -0.037                         | 0.96              |
| female       | 0.294                 | 1.34              | -0.074                         | 0.93              |
| black        | 1.071                 | 2.92              | 0.225                          | 1.25              |
| ever married | 2.187                 | 8.91              | 0.298                          | 1.35              |

- Two processes: marriage multiplies the odds of *any* child by 9 but the count among parents by 1.35; women are likelier to have any child and have slightly fewer. One Poisson averaged these.
- Fit:  $\ell$  rises from -5,383 (Poisson) to -5,228, AIC 10,778  $\rightarrow$  10,479, six more parameters. Zeros matched exactly.

## Which Count Model, and How to Decide ★

1. Start with **Poisson, robust SEs**. Coefficients are right for the mean under any variance;  $e^{\hat{\beta}}$  is the cleanest reading in the deck.
2. Run the **overdispersion test**. If it rejects and you need a likelihood (prediction, AIC, LR against a bigger model), fit the **negative binomial**.
3. Compare the **observed and predicted distribution** of counts, as in the figure. A zero problem is visible before any test.
4. **Hurdle or zero-inflated**: choose on substance. Hurdle when crossing from zero to one is a different *process* (having any children, ever being arrested). Zero-inflated when some zeros are a different *kind* of person (a nonsmoker's zero cigarettes versus a smoker's zero today). Then AIC or BIC between them; they are not nested.
5. Never compare by fit alone across models fitted to different rows.

## In the Literature: Negative Binomial ★

**Olzak (2021, *ASR*)**. Officer-involved fatalities of Black, Latino, and white residents, counted per city-year in the 170 largest U.S. cities. Many city-years have zero and the variance far exceeds the mean, so negative binomial rather than Poisson.

- One local protest against police brutality is followed by 11% fewer Black fatalities and 7% fewer Latino fatalities the next year ( $e^{\hat{\beta}} - 1$ , read as a percent), and no change for whites.
- The count model is doing what the section said: the mean is right under the Poisson too, but the standard errors would be badly too small.

## In the Literature: Excess Zeros ★

Ajrouch, Antonucci, and Webster (2016, *Journals of Gerontology: Series B*). Hours volunteered, two waves,  $n = 543$ ; most respondents report zero.

- A zero-inflated negative binomial: a logit for being a never-volunteer, and a negative binomial for hours that can also give a zero (a volunteer with no hours this wave).
- The authors read the two equations as *whether* and *how much*, which is the hurdle's reading: network proximity and contact predict whether; education predicts how much.
- Strictly, only the hurdle splits zeros from positives that cleanly. In practice the two give similar coefficients and are read the same way; they differ only in what a zero may mean, and the data rarely say which.

**Where you will see these.** Health, criminology, gerontology. In *ASR* and *AJS* a negative binomial with robust SEs usually carries the day. Use a two-part model when zero is a different kind of answer, not because zeros are many.

# Multilevel

---

## Observations Come in Groups ★

Often the question is about a group's characteristic as much as the individual's: does a Catholic school raise achievement net of who attends it (Coleman, Hoffer, and Kilgore 1982)? The intuitive move: put both kinds of variable in one OLS. High School and Beyond, 7,185 students, 160 schools:

$$\text{math}_{ij} = \beta_0 + \beta_1 \text{SES}_{ij} + \beta_2 \text{Catholic}_j + e_{ij}, \quad \hat{\beta}_2 = 1.94, \text{ SE } 0.15.$$

### What is wrong

Catholic takes 160 values, not 7,185. Students in the same school share what the model left out about it, so their errors are correlated and OLS counts each school's evidence many times over. The clustered SE (Week 2) is 0.32, twice the OLS one.

## Why Not Just OLS With Clustered Standard Errors? ★

Week 2 split the OLS error into a group shock and a personal one,  $\epsilon_{ig} = \nu_g + \eta_{ig}$ .

Same split, school as the cluster,  $u_j$  for  $\nu_g$ :

$$\text{math}_{ij} = \beta_0 + \beta_1 \text{SES}_{ij} + \beta_2 \text{Catholic}_j + u_j + e_{ij}.$$

$u_j$ : how far school  $j$  sits above or below the average school once SES and sector are accounted for, the same for every student in it;  $e_{ij}$  is the student's own deviation.

Three things to do with  $u_j$ :

- **Clustered OLS** (Week 2) used the split only to see what was wrong, then repaired the SEs without estimating  $u_j$  or its variance.
- **Fixed effects** (Week 10) estimate every  $u_j$ : 160 dummies. Catholic does not vary within a school, so it is absorbed.
- **This week** treats the  $u_j$  as draws from a distribution and estimates their *variance*. Catholic stays estimable, and the variance answers what neither of the others can: how much variation is between schools? Here 9% once SES and sector are in: Coleman et al.'s (1966) question.

## The Random-Intercept Model ★

Student  $i$  in school  $j$ , the last frame's equation with distributions:

$$\text{math}_{ij} = \beta_0 + \beta_1 \text{SES}_{ij} + \beta_2 \text{Catholic}_j + u_j + e_{ij}, \quad u_j \sim \mathcal{N}(0, \tau^2), \quad e_{ij} \sim \mathcal{N}(0, \sigma^2).$$

- Each school has its own intercept,  $\beta_0 + u_j$ : 160 draws from one distribution, not 160 constants as in fixed effects.
- $\tau^2$  is the **between-school variance**: how much school intercepts spread around  $\beta_0$ .  
 $\sigma^2$  is the **within-school variance**: how much students spread around their own school's line. Total:  $\text{Var}(\text{math}_{ij}) = \tau^2 + \sigma^2$ .
- Classmates share  $u_j$ , so  $\text{Cov}(\text{math}_{ij}, \text{math}_{i'j}) = \tau^2$ : the model *is* the correlation clustering corrects for.

**Estimation.**  $u_j$  is never observed, so each school's likelihood averages over its possible values, using the distribution assumed for it; Newton does the rest.

```
lme4::lmer(math ~ ses + catholic + (1 | school)).
```

## Two Assumptions About $u_j$ , and Why Estimation Needs Them ★

The likelihood averages over the unseen  $u_j$ . Averaging needs a distribution, and the same one for every school.

1.  $u_j \sim \mathcal{N}(0, \tau^2)$ . The distribution to average over. Mean zero is a normalization (any other mean goes into  $\beta_0$ ); normal makes the average a normal, so each school's likelihood has a closed form with one parameter,  $\tau^2$ .
2.  $u_j$  **uncorrelated with SES and Catholic**:  $E[u_j | X] = 0$ . The same  $\mathcal{N}(0, \tau^2)$  is used for high-SES and low-SES schools. If high-SES schools tended to have higher  $u_j$ , the model would credit that height to SES and  $\hat{\beta}_1$  would be too large: an omitted variable, since  $u_j$  is part of the error. Fixed effects estimate each  $u_j$  instead of averaging over it; that is what the 160 dummies buy, and why they lose Catholic.

## Reading the Fit, and the Intraclass Correlation ★

|                                 | estimate | SE   |
|---------------------------------|----------|------|
| intercept $\hat{\beta}_0$       | 11.72    | 0.23 |
| SES $\hat{\beta}_1$             | 2.38     | 0.11 |
| Catholic $\hat{\beta}_2$        | 2.10     | 0.34 |
| between schools $\hat{\tau}^2$  | 3.62     |      |
| within schools $\hat{\sigma}^2$ | 37.0     |      |

- $\hat{\beta}$ : as in any regression. A unit of SES adds 2.4 points; a Catholic school adds 2.1 at the same SES.
  - $\hat{\tau}^2$ : schools of the same sector with the same SES students still differ, SD  $\sqrt{3.62} = 1.9$  points.
  - $\hat{\sigma}^2$ : classmates differ from their school's line by SD  $\sqrt{37.0} = 6.1$  points.
- ICC =  $\hat{\tau}^2 / (\hat{\tau}^2 + \hat{\sigma}^2) = 3.62 / 40.6 = 0.09$ : of the variation SES and sector leave unexplained, 9% is between schools, 91% within. Is  $\tau^2$  zero? LR against OLS with the same  $x$ : 332.

## Predictors at Both Levels ★

Student SES (level 1) and school sector, Catholic versus public (level 2):

|                                | pooled OLS  | OLS, clustered SE | random intercept | fixed effects  |
|--------------------------------|-------------|-------------------|------------------|----------------|
| SES                            | 2.95 (0.10) | 2.95 (0.13)       | 2.38 (0.11)      | 2.19 (0.11)    |
| Catholic                       | 1.94 (0.15) | 1.94 (0.32)       | 2.10 (0.34)      | not identified |
| $\hat{\tau}^2, \hat{\sigma}^2$ |             |                   | 3.62, 37.0       |                |

- **Sector** is constant within school. Fixed effects absorb it; only a model with a between-school variance can estimate it: Catholic schools score 2.1 points higher, SE 0.34 with 160 schools as the units.
- **SES** falls from 2.95 to 2.38 to 2.19 as schools go from ignored to modelled to conditioned on: part of the pooled slope is high-SES students attending high-scoring schools.

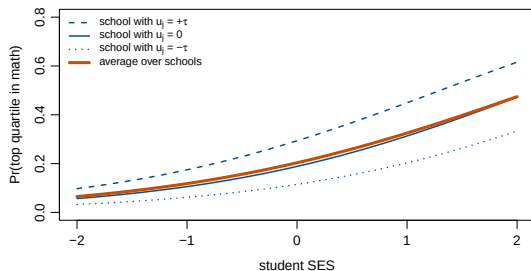
## When You Need a Random Intercept ★

1. **The split itself is the finding.** Tomaskovic-Devey et al. (2020, *PNAS*): earnings in 14 high-income countries over 25 years, two billion job-years nested in 50 million workplace-years, split into between-workplace and within-workplace variance. The between-workplace share rose in 12 of 14 countries and fell in none, fastest in the private sector. That share is  $\tau^2/(\tau^2 + \sigma^2)$  with the workplace as the group; fixed effects and clustered OLS have no such number.
2. **A group-level variable with few groups.** Clustered SEs are a many-clusters tool: they rely on the clusters being many independent draws and are unreliable below a few dozen (Cameron and Miller 2015). Twelve countries or twenty firms: the random intercept's model-based SE is the credible one, at the price of assuming  $u_j$  normal and uncorrelated with  $x$ . With 160 schools either works: Catholic 1.94 (0.32) clustered, 2.10 (0.34) random intercept.

**Otherwise:** a group-level coefficient, many groups, no interest in the variance. OLS with clustered SEs is enough.

## Random-Intercept Logit, and R ★

Top quartile in math:  $\Pr(y_{ij} = 1) = \Lambda(\beta_0 + \beta_1 \text{SES}_{ij} + \beta_2 \text{Catholic}_j + u_j)$ ,  
 $u_j \sim \mathcal{N}(0, \tau^2)$ .



|                | $\hat{\beta}$ | SE    |
|----------------|---------------|-------|
| SES            | 0.674         | 0.045 |
| Catholic       | 0.391         | 0.112 |
| $\hat{\tau}^2$ | 0.34          |       |

- Public schools at  $\text{SES} = 0$ : 0.12 one SD below the average school, 0.19 at  $u_j = 0$ , 0.29 one SD above; 0.20 averaged over schools.  $\hat{\beta}_1$  is one school's slope; the average is the curve for students overall. Report the one the question asks.
- No closed form for the average over  $u_j$ : `glmer(..., family = binomial, nAGQ = 10)` does it on a grid.

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